



CLINICAL PROGRAMMING BOOTCAMP • MODULE 02

Clinical Trials 101

How a trial produces data

Before you can map clinical data, you need to know where it comes from — who runs a trial, what happens to a patient, and how every measurement gets recorded.

No clinical background assumed. Every term defined on first use.

By the end of this module you can...

1

Explain what a clinical trial is and why medicines must be tested in people before approval.

2

Name the phases and say what each phase is trying to find out.

3

Identify who does what sponsor, CRO, investigator, site staff, ethics committee, regulator.

4

Follow a subject's journey from screening and consent through treatment to end of study.

5

Point to where data is born the visit, the CRF, and the EDC system.

What is a clinical trial?

A research study in people

A clinical trial is a carefully controlled study that tests whether a medical treatment is safe and whether it works — in human volunteers, following a written plan agreed in advance.

Nothing is decided casually: who can join, what is measured, when it is measured, and how results are analyzed are all fixed before the first patient enrolls.

Is it safe?

What side effects occur, how often, how serious?

Does it work?

Does it improve the outcome it is meant to improve?

At what dose?

How much, how often, for how long?

For whom?

Which patients benefit — and which should avoid it?

A trial is one step in a long road



Typically 10-15 years and a very high failure rate — most compounds never reach patients. Durations shown are typical, not rules.

The four phases



Participant numbers are typical ranges and vary widely by therapeutic area.

The protocol: the study's single source of truth

Every trial is run according to a protocol — a formal written document approved before enrollment starts. If you ever wonder “why is the data like this?”, the answer is usually in the protocol.

1

Objectives

What the study is trying to find out

2

Design

Randomized? Blinded? How many arms? How long?

3

Eligibility

Inclusion / exclusion criteria — who may join

4

Schedule of assessments

Which measurements happen at which visit

5

Endpoints

The primary and secondary outcomes that decide success

6

Statistical methods

How the results will be analyzed — fixed in advance

The Schedule of Assessments is a programmer's best friend — it tells you exactly what data to expect, and when.

Who's who in a clinical trial

1 Sponsor

Owns and funds the trial — usually a pharma or biotech company.

2 CRO

Contract Research Organization — hired by the sponsor to run parts of the trial.

3 Investigator (PI)

The doctor responsible for the trial at a site. PI = Principal Investigator.

4 Site staff

Study coordinators and nurses who see subjects and enter the data.

5 Ethics committee

IRB / IEC — independent board that protects participants' rights and safety.

6 Regulator

FDA, EMA, PMDA — approves trials and reviews the final submission.

Clinical programmers usually sit with the sponsor or the CRO — downstream of the site, upstream of the regulator.

Sites and subjects

Site

A hospital or clinic where the trial is actually carried out.

One trial usually runs at many sites, often in several countries.
Each site has an ID (e.g. 01, 02) and its own staff.

More sites = faster enrollment, but more variation in how data gets recorded.

Subject

One participant in the trial. Also called a patient or participant.

Each subject gets a number that is unique within their site — which is exactly why SDTM needs USUBJID, an identifier unique across the whole study.

Subject 001 at Site 01 $\neq$ Subject 001 at Site 02

Two different people. This is the #1 identifier trap.

What actually happens to a participant



Informed consent comes first. Nothing may be done for research purposes until the subject has voluntarily agreed, in writing, after being told what the study involves.

Randomization, control, and blinding

Randomization

Subjects are assigned to treatment groups by chance, not by choice.

Why it matters: Stops conscious or unconscious selection from skewing the groups.

Control arm

A comparison group — placebo (an inactive look-alike) or an existing standard treatment.

Why it matters: Without a comparison you cannot tell whether the drug caused the change.

Blinding

Hiding who got what. Single-blind: subject doesn't know. Double-blind: subject and investigator don't know.

Why it matters: Expectations change how people report and assess symptoms.

“Arm” = one treatment group. In our mock study ABC-01 there are two arms: Drug A and Placebo.

What actually gets collected

At each visit, specific things are measured or asked. Each becomes a category of data — and later, an SDTM domain.

Who the subject is	Age, sex, race, country, treatment arm	DM
What happened to them	Side effects and medical problems	AE, MH
What they were given	Study drug doses; other medications taken	EX, CM
What was measured	Blood pressure, labs, ECGs, physical exam	VS, LB, EG, PE
How the study ended	Completed or withdrew, and why	DS

You already met these two-letter codes in Module 01 — this is where they come from.

The CRF: how a measurement becomes data

Case Report Form (CRF)

The form used to record the information the protocol requires for each subject.

One form per topic — a demographics form, an adverse event form, a vital signs form. Historically paper; today almost always electronic (eCRF) inside an EDC system.

One CRF form usually becomes one raw dataset — which you then map to a domain.

VITAL SIGNS

Visit

SCREENING

Date

20-FEB-2024

Systolic BP

122 mmHg

Diastolic BP

80 mmHg

Pulse

68 beats/min

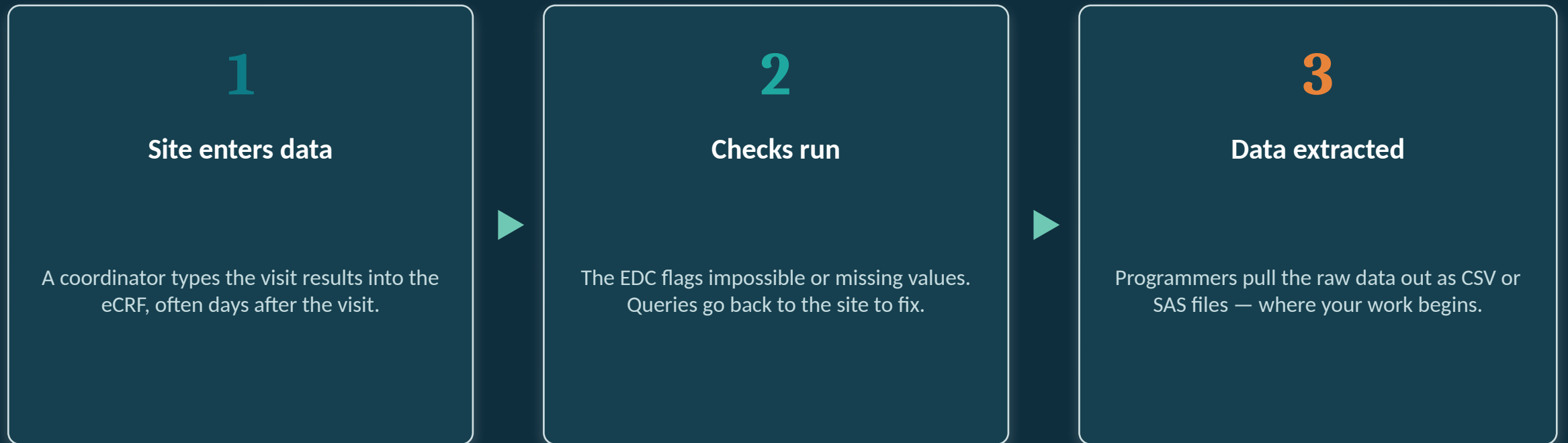
Temperature

36.7 °C

Weight

70.5 kg

From form to database: EDC



EDC = Electronic Data Capture. Data cleaning is continuous — the database is “locked” only when the study is complete.

Trial vocabulary you'll hear daily

Protocol The study's written plan	Site A hospital/clinic running the trial	Subject One participant
Arm A treatment group (e.g. Drug A)	Visit A scheduled appointment	Baseline The pre-treatment reference value
Endpoint The outcome that decides success	Adverse event Any untoward medical occurrence	Eligibility Inclusion / exclusion criteria
Query A question raised about a data value	Database lock The final freeze of the data	GCP Good Clinical Practice — the ethical standard

From the visit to the regulator



Module 01 showed you the standards. Module 02 showed you where the data comes from. From Module 03 on, you build it yourself.

CLEAR THESE UP NOW

Common misconceptions



“The data arrives clean.”

It doesn't. Sites differ, formats differ, and queries are still open when you start.



“Subject numbers are unique.”

Only within a site. Study-wide uniqueness is something you construct.



“Missing means zero.”

Missing means not collected or not done. Never invent a value to fill a gap.



“We can change what we measure.”

The protocol fixes it in advance. Deviations are documented, not silently fixed.



“Baseline is just the first record.”

Baseline is defined by the protocol — usually the last value before first dose.



“Programmers don't need clinical knowledge.”

You must know what a value means to judge whether your mapping is right.

Glossary of acronyms

CRO	Contract Research Organization	PI	Principal Investigator
IRB / IEC	Institutional Review Board / Independent Ethics Committee	GCP	Good Clinical Practice (ICH E6)
ICH	International Council for Harmonisation	CRF / eCRF	Case Report Form (electronic)
EDC	Electronic Data Capture	SoA	Schedule of Assessments
AE	Adverse Event	FDA / EMA / PMDA	US / European / Japanese regulators
TLF	Tables, Listings and Figures	Database lock	Final freeze of the trial data

WHAT'S NEXT

You know the story — now learn the tools

1

Module 03 Your toolkit: SAS & R side by side

2

Notebook 01 SAS basics — DATA steps and PROCs

3

Notebook 01 (R) R & tidyverse basics — the same skills

4

Then Reading the raw ABC-01 data and building your first domain

Keep the mock study ABC-01 open — from here on, everything you build uses it.